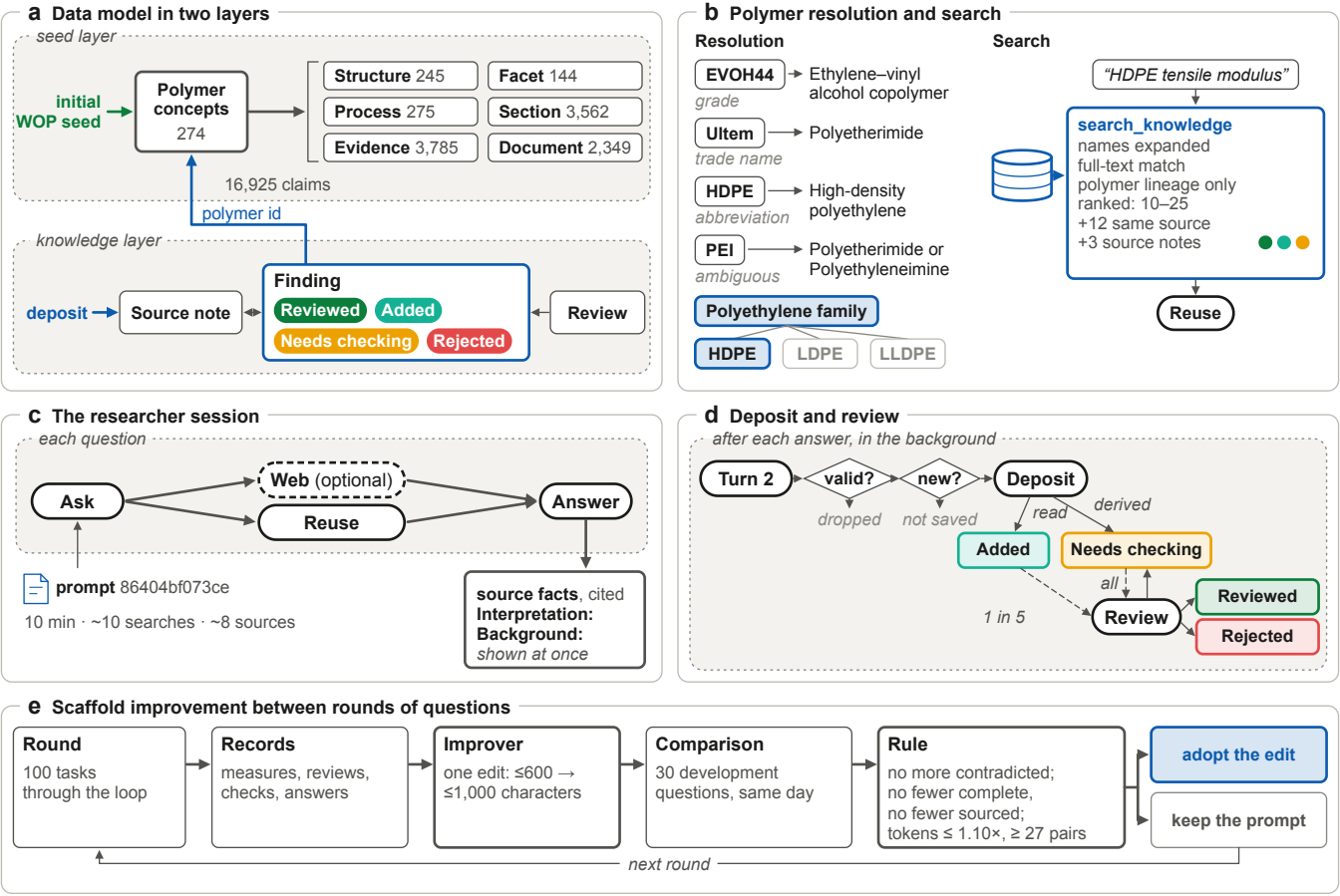

a Data model in two layers
seed layer
initial WOP seed
Polymer concepts 274
Structure 245
Facet 144
Process 275
Section 3,562
Evidence 3,785
Document 2,349
16,925 claims
polymer id
knowledge layer
deposit
Source note
Finding
Reviewed
Added
Needs checking
Rejected
Review
b Polymer resolution and search
Resolution
EVOH44
grade
Ethylene–vinyl alcohol copolymer
Ultem
trade name
Polyetherimide
HDPE
abbreviation
High-density polyethylene
PEI
ambiguous
Polyetherimide or Polyetheryleneimine
Polyethylene family
HDPE
LDPE
LLDPE
Search
"HDPE tensile modulus"
search_knowledge
names expanded
full-text match
polymer lineage only
ranked: 10–25
+12 same source
+3 source notes
Reuse
c The researcher session
each question
Ask
Web (optional)
Reuse
Answer
prompt 86404bf073ce
10 min · ~10 searches · ~8 sources
source facts, cited
Interpretation:
Background:
shown at once
d Deposit and review
after each answer, in the background
Turn 2
valid?
new?
Deposit
dropped
not saved
read
derived
Added
Needs checking
all
1 in 5
Review
Reviewed
Rejected
e Scaffold improvement between rounds of questions
Round
100 tasks through the loop
Records
measures, reviews, checks, answers
Improver
one edit: ≤600 → ≤1,000 characters
Comparison
30 development questions, same day
Rule
no more contradicted;
no fewer complete,
no fewer sourced;
tokens ≤ 1.10×, ≥ 27 pairs
adopt the edit
keep the prompt
next round